

Accelerating the Development of Machine Voice User Interfaces with Immersive Environments

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ABSTRACT

This paper proposes a conceptual framework leveraging VR technologies to develop large language model (LLM)-based Voice User Interfaces (VUIs) for industrial applications, addressing key adoption challenges. By integrating virtual twins and immersive environments, the framework supports essential development steps, including data collection, LLM training, usability testing, and iterative design validation. The study shows how VR enables realistic simulations, faster development, and reduces the dependency on physical prototypes in the VUI development, offering improvements in efficiency and user-centered design. Initial findings of the proof-of-concept study highlight the framework's potential to address industrial requirements.

Index Terms: Voice User Interfaces, Virtual Reality, Immersive Environments, Industrial Applications, Human-Machine Interaction, Large Language Models, Technology Acceptance Model

1 INTRODUCTION, MOTIVATION & RELATED WORKS

VUI offer significant benefits for industrial applications, enabling hands-free, intuitive interaction in complex and safety-critical environments. These systems improve operational efficiency, reduce cognitive load, and enhance user accessibility. However, their adoption remains limited due to complex demands, such as data privacy, operational safety, and usability concerns.

Recent advancements in Artificial Intelligence (AI) technologies address these challenges by enabling VUIs to manage dynamic interactions, adapt to user needs, and improve performance. Despite this, significant obstacles exist, especially in data collection for training algorithms, usability testing, and optimization. LLMs are transformer-based neural networks, pre-trained on extensive text data to enable advanced language understanding and responses [10].

Our research aims to develop a universal comprehensive platform for AI-based voice control of industrial machines and automated systems, as currently existing solutions often focus on single implementation steps. This approach benefits specialized machine manufacturers with limited production volumes by integrating VR technologies, including virtual and digital twins, to accelerate development and enhance system usability [2].

Virtual twins and VR environments (VREs) could play a significant role in VUI development. For example, Ludwig et al. highlight the role of VREs in streamlining manufacturing logistics and cognitive and physical demands [7]. Norda et al. demonstrate their effective-

ness in accelerating operator training [11]. VR-based simulations also reduce the reliance on physical prototypes through front-loading, allowing for parallel software and hardware development [3]. This decoupling minimizes development time and facilitates early issue identification [4].

By combining VUIs with VREs throughout the product lifecycle, the conceptual framework we propose facilitates comprehensive design, testing, and continuous optimization (see Fig. 1). It addresses key industrial challenges, ensuring VUIs deliver robust, user-centric solutions with maximum value and usability.

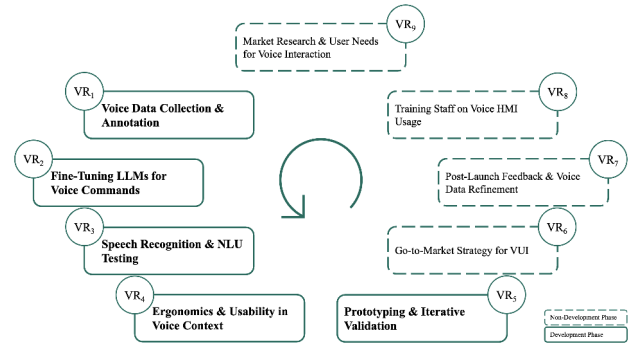


Figure 1: Development phases and VR Integration

2 CONCEPTUAL FRAMEWORK ON VUI DEVELOPMENT ACCELERATION THROUGH VIRTUAL REALITY TECHNOLOGY

The framework is divided into developmental and non-developmental phases, as shown in Fig.1, with our research focusing on the integration of VR into the developmental stages of the LLM-based VUI.

In the Voice Data Collection and Annotation (VR₁) phase, VR provides an immersive and controlled environment for collecting realistic voice interaction data. By simulating user interactions within a virtual prototype of the machine, VR ensures that datasets capture naturalistic speech patterns, environmental noise variations, and contextual nuances that would occur in real-world usage.

During the Fine-Tuning of LLMs for Voice Commands (VR₂), VR offers a platform to train and validate models under realistic conditions. This step benefits from VR's ability to replicate environmental contexts and operational scenarios, allowing LLMs to adapt to diverse inputs and improve command recognition accuracy.

In the Speech Recognition and Natural Language Understanding (NLU) Testing (VR₃) phase, VR enables developers to test and refine these components by simulating complex, context-rich environments. This ensures that the VUI performs robustly even in challenging conditions, such as noisy industrial settings or multi-user scenarios, improving both speech recognition accuracy and contextual interpretation.

The Ergonomics and Usability Testing in Voice Contexts (VR₄) phase leverages VR to study how users physically and cognitively

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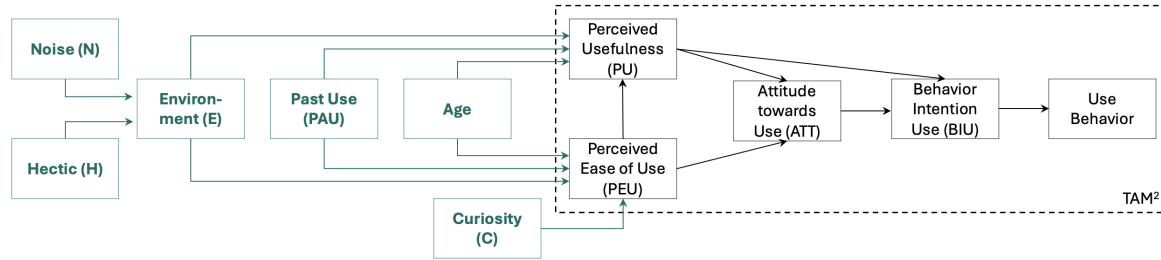


Figure 2: Technologies Acceptance Model for VUIs

interact with the VUI. VR facilitates detailed analysis of user behavior, interface accessibility, and interaction satisfaction, providing critical insights into how the system can be optimized for user-centered design.

Finally, during Prototyping and Iterative Validation (VR₅), VR allows developers to iteratively refine the VUI by testing different designs and configurations in a risk-free, flexible immersive environment. This not only accelerates the prototyping process but also reduces costs and resource requirements compared to traditional physical prototypes.

By integrating VR into these key development steps, the efficiency, precision, and user-centeredness of VUI development can be significantly enhanced, laying the foundation for robust and adaptable voice interaction systems.

3 IMPLEMENTATION

To validate the developed framework, several use cases were involved in the project. A virtual prototype of a vending machine as well as a virtual twin of a CNC machine were implemented in the VR engine PolyVR [6], using product planning data (CAD, ECAD, machine programming) as the foundation. Those prototypes were optimized for deployment in VR to enable seamless simulation and enhanced with detailed texturing for visual realism. Integration of the VUI with the VR applications was achieved through a combination of UDP for the real-time exchange of audio packets and REST interfaces for machine status updates, notifications and execution of command requests. The result was a robust and dynamic testing environment in which voice commands could be accurately validated in a realistic and interactive virtual context, providing critical insights into system usability, performance, and user interaction.

For the first three steps — Voice Data Collection, Fine-Tuning LLMs for Voice Commands, and Speech Recognition & NLU Testing — VR was used to create an immersive environment that enhanced data quality, testing precision, and user interaction realism. By applying the Wizard-of-Oz (WoZ) method [9] for advanced data collection, we were able to generate realistic and large datasets for training the VUI's LLM. This approach enabled systematic collection of high-quality interaction data, capturing naturalistic spoken exchanges between the machine and users, thereby providing valuable insights into user behavior and dialogue patterns essential for refining the VUI's performance [5].

The final two steps of the development phase —Ergonomics & Usability in Voice Context and Prototyping & Iterative Validation — leveraged VRE's to gain a deeper understanding of user interactions with the machine, their satisfaction with the interface, and the influence of the environment in which the machine is embedded. To achieve this, we developed a customized variation of the TAM [1], inspired by the work of Manis and Choi [8] (see Fig. 2). The VRE provided a controllable environment to systematically test both the machine's surroundings and associated TAM variables, enabling simulations of diverse conditions like noise, pace, or complexity to develop robust, adaptable products for various customer environments.

4 VALIDATION & NEXT STEPS

All steps of the development phase were tested in a proof-of-concept study, which simultaneously involved the collection of datasets and application of the customized TAM. This initial study demonstrated significant potential and promise for the integration of VR in the development of VUIs, highlighting the effectiveness in data generation and user evaluation. The next steps involve conducting a study with more participants to perform a statistical analysis of the TAM. Additionally, further refinement and development of the virtual prototypes will focus particularly on the machine's environmental context, which will deepen the understanding of how the environment influences user perceptions and interactions, enhancing the design and usability of the VUI.

ACKNOWLEDGMENTS

This research is funded by the German Federal Ministry for Education and Research (BMBF) within the project "DialoKIM: Dialog-oriented artificial intelligence for intuitive machine control". This work will be not possible without the cooperation with our project and associated partners [2].

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